

The Hong Kong Polytechnic University

Department of Computing

COMP4913 Capstone Project

Report (Interim / Final*)

Intelligent Property Management System with LLM-based Autonomous
Agents

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Abstract

The property management industry faces increasing challenges in addressing tenants' needs, managing maintenance efficiently, and handling the unique complexities of multiple properties. Traditional systems often suffer from poor communication, limited automation, and high operational costs, resulting in delayed responses and inconsistent service. To overcome these issues, this project proposes the development of an Intelligent Property Management System (IPMS) powered by Large Language Models (LLMs) and autonomous agents using the LangGraph framework.

The system integrates LLMs such as DeepSeek to facilitate natural, conversational interactions with tenants. Three specialized autonomous agents are developed: a Communication Agent that handles tenant queries and provides appropriate responses; a Task Allocation Agent that assigns maintenance tasks and notifies relevant stakeholders; and a Monitoring Agent that tracks task progress and updates property status in real-time. This multi-agent architecture is designed to be modular, scalable, and aligned with agile development principles.

To evaluate system performance, a controlled experiment was conducted. A rubric was created with key criteria—Accuracy of Response, Completeness of Information, Clarity and Readability, Response Time, and Relevance of Context. A dataset of 100 realistic property owner queries was compiled, with benchmark answers manually defined. The experiment compared responses from the proposed multi-agent system and DeepSeek alone. Results showed that the multi-agent system achieved an average score of 84.32, significantly outperforming DeepSeek's 76.77. This project demonstrates a practical and innovative application of AI to modernize property management. By combining LLMs with structured autonomous agents, the IPMS delivers improved responsiveness, reduced workload, and greater tenant satisfaction—providing a timely and scalable solution for the industry.

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Chapter 1 INTRODUCTION

1.1 Background of Study

The property management sector is under increasing pressure as tenant expectations grow and the challenges of managing a vast array of properties become more complex. Traditional property management systems—heavily reliant on paper-based workflows and fragmented communication channels—are becoming less effective. These outdated systems are prone to errors and inefficiencies, often resulting in delays in responding to tenant inquiries, poorly coordinated maintenance activities, and a lack of real-time monitoring of property conditions.

For example, in a large mixed-use residential complex with hundreds of units, tenants frequently report maintenance issues like plumbing leaks, electrical failures, or heating malfunctions. In a conventional system, tenants would call a receptionist or fill out a paper form to log the issue, which is then manually processed by the property management team. This process can take several hours or even days, resulting in an unaddressed issue and a growing risk of property damage. For instance, water leaks can lead to severe water damage, increased repair costs, and, most importantly, tenant dissatisfaction. Additionally, because of the lack of automation, the property management team often fails to prioritize urgent issues effectively, leading to non-critical tasks being handled before more pressing concerns. This inefficiency not only prolongs problem resolution but also increases operational costs as delays accumulate.

As tenant expectations evolve towards quicker, more transparent communication and efficient service, property managers are turning to technology to meet these demands. Artificial intelligence (AI) has emerged as a promising solution, offering significant improvements over traditional methods. Specifically, Large Language Models (LLMs) and multi-agent systems are being explored to automate various aspects of property management, such as tenant communication, maintenance request prioritization, and real-time property monitoring.

AI-driven tools can address some of the critical gaps in current property management systems. For instance, AI-powered chatbots and virtual assistants can manage tenant inquiries, reducing wait times for responses and allowing property managers to handle higher volumes of requests. Autonomous agents can also prioritize and assign maintenance tasks more efficiently, ensuring that critical repairs are addressed in real time, reducing delays and preventing costly emergency repairs. Furthermore, AI can

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enable predictive maintenance by continuously monitoring property conditions—such as energy consumption, security systems, and appliance performance—and alerting managers to issues before they escalate.

According to Market Research Future (2023), the digital property management market is projected to grow at an annual rate of over 5%, with AI-based solutions driving much of this growth. This shift is essential to keep up with the increasing demand for streamlined operations, improved tenant satisfaction, and reduced operational costs.

The success of an Intelligent Property Management System (IPMS) hinges on the active participation and support of several key stakeholders:

- **Property Managers:** Property managers are responsible for overseeing the daily operations of properties, from tenant relations to facility maintenance. They will benefit from the AI-driven system by having more efficient workflows, improved tenant communication, and real-time data to make better-informed decisions. The automated allocation of maintenance tasks will also reduce their administrative burden, allowing them to focus on more critical management tasks.
- **Tenants:** Tenants are the end-users of the property management system and will experience direct improvements in the form of faster response times, better communication, and more reliable maintenance services. An AI-powered system that handles their inquiries and maintenance requests can significantly enhance their overall satisfaction, leading to better tenant retention rates.
- **Property Owners:** Property owners are interested in the profitability and operational efficiency of their properties. The implementation of an AI-powered property management system will lead to reduced costs, increased tenant retention, and improved property conditions. This will result in higher property value, greater return on investment (ROI), and long-term operational savings.
- **Maintenance Teams:** Maintenance teams are responsible for carrying out repairs and handling maintenance requests. By automating task allocation and prioritization, the system will help maintenance staff address issues faster and more efficiently. The use of predictive maintenance will also help identify potential problems before they become critical, reducing the frequency of emergency repairs and lowering overall maintenance costs.

1.2 Problem Statement

Current property management systems face several key issues:

- **Manual Data Handling and Inefficient Communication:** Most tenant requests, complaints, and maintenance requests are handled through paper-based systems, therefore, taking longer to respond, not to mention the unsatisfied tenants. The receptionists are overwhelmed with communication tasks, especially when it is related to many properties that are managed.
- **Limited Automation:** Some systems are simply designed to give reminders for payments or scheduling of maintenance among others but they are not integrated between the different functions. This lack of coordination undermines efficiency and creates redundancies.
- **Inconsistent Maintenance and Monitoring:** Maintenance requests are typically delayed due to manual task allocation, and there is no proactive system for tracking property conditions to prevent breakdowns or costly emergency repairs.

1.3 Objectives

This project aims to create an Intelligent Property Management System (IPMS) that is powered by Large Language Model (LLM) based Autonomous Agents that will bring a new paradigm for how property management tasks are performed. Key areas like tenant communication, maintenance request management and real time property monitoring will be automated, while improving operational efficiency and tenant satisfaction.

The specific, measurable outcomes of this project include:

- **Automated Tenant Interaction:** Create a communication system to respond to tenant inquiries, complaints and maintenance requests by tenant using natural language processing (NLP) in text and voice using LLM. The system will cut the response time to tenant inquiries in half versus manual methods.
- **Smart Maintenance Task Allocation:** Designing autonomous agents which prioritize and assign maintenance requests by urgency, such that critical tasks are handled in real time. Expected outcome: On average, maintenance response times are reduced by 40% and task mismanagement incidents are reduced.
- **Real-Time Property Monitoring and Reporting:** Continuous monitoring of the property conditions (for example energy usage, security) through autonomous agents that detect the issues and notify property managers in real time. By enabling predictive maintenance, it is expected to reduce the emergency maintenance occurrences by 30%.

- **Improved Tenant Satisfaction and Reduced Costs:** The system is designed to ensure streamlining of communication, task allocation and monitoring processes, that will result in improved tenant satisfaction and 25% reduction in overall operational cost for the property managers.

The project will contribute to the field of property management by: An introduction of an AI driven approach for automating repetitive work, a scalable solution that can be adapted to different sizes and types of properties, and how the use of multi-agent systems can improve the efficiency of complex, multi-faceted operations over single agent ones.

1.4 Scope of Project

The scope of this project includes the development of an AI-driven property management system comprising the following key modules:

- **Tenant Interaction Module:** Facilitates communication through LLM-powered natural language processing (NLP) to address tenant inquiries and requests.
- **Task Allocation Module:** Utilizes autonomous agents to manage and prioritize maintenance tasks.
- **Monitoring and Reporting Module:** Monitors the real condition of the property and produces reports that will be beneficial for the property managers.
- **Integration Capabilities:** Can easily be integrated with other current property management applications and even smart apps for increased efficiency.

The concepts underlying the design of the system are scalability, efficiency, and modularity, thus making it possible to apply the system in both small- and large-scale property management.

1.5 Relevance of the Project

In this regard, this project has significant applicability in responding to the changing needs of the property management market. By leveraging advanced AI technologies such as LLMs and multi-agent systems, it aims to deliver the following benefits:

- **Improved Efficiency:** Time and again, there will be a decrease in operational costs due to integration of automation in performing of certain tasks.
- **Enhanced Tenant Experience:** Better and quicker response and feedback will make the tenants happy.
- **Scalability:** The structure of the system is modular and allows expanding it as the number of tenants and properties continues to rise.

- **Market Competitiveness:** With the growing market need for AI in property management tools, this project will enable property managers to stand ready for competition in the increasing market trends.

1.6 Project Feasibility

The feasibility of this project is supported by several factors:

- **Technical Feasibility:** The current presence of sophisticated AI instruments such as DeepSeek API guarantees that the LLM component of the system can be created efficiently. Furthermore, there are ready solutions for multi-agent systems as Celery for task management in the form of frameworks and libraries.
- **Economic Feasibility:** As more properties owners start to show interest and willingness to invest in artificial intelligence-based property management tools, the project is poised to meet the emerging market needs, and therefore presents a good proposition for investors.
- **Operational Feasibility:** The system is designed as a set of modules, and each module can be developed separately, with defined deliverables for each, which minimizes risks and allows completing the work on time.

Compared with the existing systems, this project has solved the existing problems and utilized the opportunities of AI technologies, which is a milestone in the field of property management.

Chapter 2 LITERATURE REVIEW

In this project, I looked at the possibility of deploying artificial intelligence and application of autonomous agents in property management and the use of the existing property management systems. In this review, we look at how the current technologies are used, where they lack, and how advanced technologies such as LLMs (Large Language Models) and multi agent systems can improve upon these limitations.

Property management systems (PMS) are intended to help and automate the tasks associated with managing properties (Azman & Ishak 2024). These are tenant communication, maintenance requests, rent collection and facility management. There has been significant adoption of current PMS solutions such as Yardi and AppFolio in the industry (Bouchrika, 2024). However, these systems are often limited by:

- Manual input and process inefficiency: However, many systems continue to depend heavily on human intervention for communication with tenants, resolution of issues, and data entry, resulting in delays and human errors (Malloy et al., 2023).
- Limited automation: In fact, some PMS solutions offer automation for rent collection and maintenance tracking, but they still lack powerful AI automation for tenant interactions and decision making (Roy et al., 2023).
- Lack of contextual communication: A context aware understanding of tenant requests is difficult for existing systems to accomplish, resulting in frustration and miscommunication (Babakian, 2024).

An amazing thing has happened in the property management industry, and that's artificial intelligence has come in to not only automate jobs but also perform them with a high degree of accuracy. Although LLMs and autonomous agents have only recently started to augment PMS systems, they still have a way to go. Chatbots that deploy NLP to automate tenant communication, such as Zillow and RealPage, are being used to make this happen. But these systems can answer basic questions about rent, property features, and availability (Brodsky, 2023). Unfortunately, these solutions do not provide the nuanced, context sensitive conversations that could truly improve user experience (Babakian, 2024). Successfully, multi agent systems (MAS) have been used in other domains such as healthcare, logistics, robotics. Since a PMS consists of many tasks, which would need to be executed based on priority and resource availability, multi agent frameworks can help in distributing tasks. Given that these systems enable agents to act

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separately, interact, and adapt to changes, they improve scalability and flexibility in property management (Maldonado et al., 2024).

In addition, traditional PMS solutions are found to scale poorly when the number of properties and tenants continue to grow in areas like communication, maintenance management and payment processing. This challenge can be addressed by multiagent systems which can distribute the task to multiple agents, thus they do not overburden a single agent while scaling efficiently. With this increasing digitization, there is a huge concern about the security of tenant and property data. Compliance with data privacy regulations such as GDPR but still retaining efficient data usage is a big challenge (Labadie & Legner, 2023).

Integration of LLMs with multi agent systems (MAS) is an important contribution of this project. While an individual agent may be capable of handling specific tasks, multi-agent systems offer a more robust approach by:

- Distributing workload: Multiple agents can be used to manage task allocation based on each of them specialized and the resources available. This approach avoids the bottleneck of an overloaded single agent.
- Contextual awareness: Integrating LLMs with autonomous agents will allow tenants to have more human like, context aware interactions with the tenants, thereby increasing tenant satisfaction. For instance, the system can be taught to 'recall' previous conversations if a tenant mentions something that was brought up before and respond with personalized, relevant answers.
- Real-time decision-making: Dynamic adaptation to changes in the environment and real time decision making are possible in multi agent systems. For instance, a maintenance agent may allocate work to repair persons who are available and nearby to the site, and urgent.

Chapter 3 METHODOLOGY

3.1 Research Methodology

The identified challenges in property management are addressed in this project through technical development and system evaluation using mixed methodology. In its method, the research applies AI technologies, including Large Language Models (LLMs) and autonomous agents, synthesizing theoretical investigation with practice. Key research methodologies include:

- Literature Review: Analysis of current AI-enabled property management solutions, multi-agent systems, and LLM applications.
- Market Analysis: Identification of gaps and opportunities within existing property management systems through industry reports and user feedback.
- System Prototyping: Iterative development and testing of the Intelligent Property Management System to ensure alignment with project objectives.
- Evaluation Metrics: Measuring system performance in areas such as Accuracy of Response, Completeness of Information, Clarity and Readability, Response Time, and Relevance of Context.

3.2 Software Development Life Cycle (SDLC)

The project uses an agile Software Development Life Cycle (SDLC) with iterative and incremental development cycles. Such a methodology allows for a dynamic work process because testing continually and feedback is constant to make sure the system is properly fulfilling the project goals. Each phase focuses on one aspect of the system while keeping the flexibility to implement feedback from stakeholders and adjust characteristics of the system based on new needs. The above approach makes it easy to involve all stakeholders in the development of the system while reducing risks and increasing the efficiency of the enhancements that are made during the development of the system.

3.3 Project Phases and Activities

Stage 1: System Architecture and Design

In this stage the architecture of the system will be developed having modular architecture that consist of the autonomous agents. LLMs provide tenants with a natural language understanding which allows them to chat with their community, either through voice or text. LLMs keep track of this context, and provide personalized and relevant responses. Our system integrates autonomous agents to provide efficient management of diverse functionalities.

- Task Allocation Agents: Choose a maintenance task depending on the urgency.
- Communication Agents: Keep track of tenant inquiries and update them on time.
- Monitoring Agents: Alert management to property conditions that are continuously checked for energy inefficiency, security breaches, theft, etc.

I will then present a flowchart showing an interaction process of each agent and their role. This will help us better understanding how our multi agent system works.

Communication Agent Flowchart: This agent handles tenant inquiries, processes the context, fetches applicable information, and responds to tenants; escalates issues as needed.

Task Allocation Agent Flowchart: This agent receives the maintenance requests and determines the urgency of each, then assigns the task to whom it belongs, notifies the tenants on the status and updates the task completed.

Monitoring Agent Flowchart: This agent is continuously monitoring the property condition, detecting anomalies or issues, notifying management in case of critical issues, suggesting preventive measures and reporting on the maintenance.

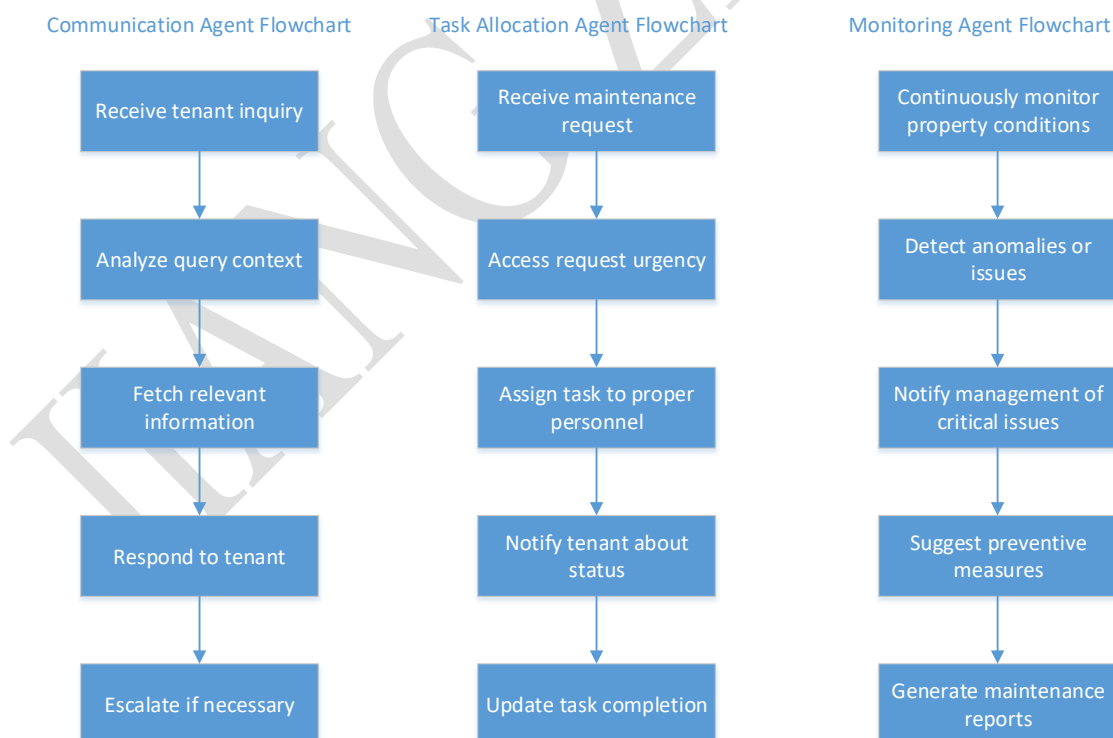


Figure 1. Agent Flowchart

A detailed flowchart of all interaction between the agents in the Intelligent Property Management System is shown here. We illustrate how each agent occurs, what they

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Tenant: A chat interface is used by Tenant to initiate requests. They can ask you for maintenance issues or inquire about billing.

Communication Agent: Gets requests from the Tenant. Calls the Task Allocation Agent with the request for further action.

Task Allocation Agent: The request is evaluated and appropriate Maintenance Agent is assigned tasks based on nature and urgency of the request.

Maintenance Agent: Assigns the tasks to handle, such as scheduling repairs or taking care of maintenance needs. When the task is done, this agent sends updates back to the Communication Agent.

Monitoring Agent: It keeps checking property conditions, such as energy usage and security status. It reports any major findings or problems back to the Task Allocation Agent in order to proactively manage.

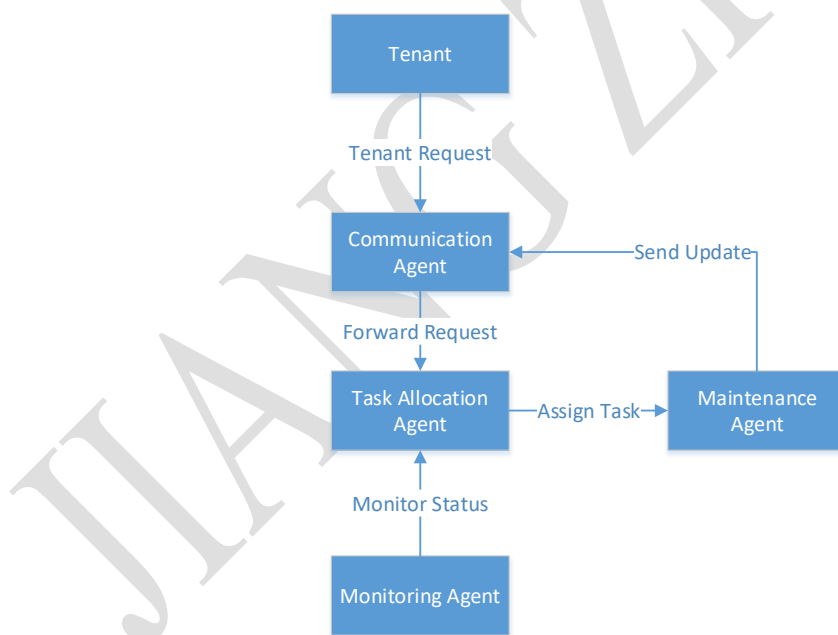


Figure 2. A detailed flowchart of all interaction between the agents

Stage 2: Development and Implementation

The second stage focuses on the development and implementation of the system using appropriate frameworks and programming languages. The system features dashboards tailored for property managers to track activities, view reports, and manage maintenance requests efficiently. The backend was developed using a combination of Express.js and

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Flask, where Express.js will handle API endpoints for user authentication, property retrieval, and database interactions, while Flask will be responsible for managing interactions between the LLM, autonomous agents, and task delegation workflows.

To enhance tenant communication, the system will integrate DeepSeek API, enabling natural language processing for automated query resolution, classification of maintenance requests, and providing property insights. Additionally, a set of autonomous agents will be implemented to manage different operational tasks. The Task Allocation Agent will assign maintenance jobs, the Communication Agent will handle tenant and staff interactions, and the Monitoring Agent will track real-time status updates.

The system's database integration will be managed through SQLite for local development, and streamline database operations. The database will store information related to users, properties, rental agreements, and maintenance activities, ensuring data is consistently available to all system components.

A key enhancement in this stage is the integration of LangGraph, which will facilitate structured workflows between autonomous agents, ensuring efficient task delegation and real-time coordination. LangGraph will direct tenant requests to the appropriate agents based on the nature of the query—routing maintenance-related issues to the Task Allocation Agent and general inquiries to the AI-powered Communication Agent. The Monitoring Agent will oversee task progress and dynamically adjust workflows, such as reassigning delayed maintenance tasks or escalating unresolved issues to management. This integration will ensure seamless multi-agent collaboration while preventing inefficiencies like deadlocks or redundant task execution.

Stage 3: Testing and Evaluation

Upon completion of development, the system will undergo a structured and rigorous testing phase to validate its functional performance, efficiency, and real-world applicability. Testing will be divided into two phases: modular testing and integrated system testing.

- Modular Testing

Each of the three autonomous agents will first be tested independently:

The Communication Agent will be evaluated based on its ability to interpret tenant queries and provide relevant, context-aware responses.

The Task Allocation Agent will be assessed for how effectively it assigns maintenance requests to appropriate personnel, and how well it communicates status updates to owners and tenants.

The Monitoring Agent will be tested on its ability to track task progress, update statuses in real time, and detect any anomalies or delays in task execution.

- Integrated Testing with Multi-Agent Collaboration

Following module-level testing, the full multi-agent system will be tested in a real-time simulated environment to assess the performance and efficiency of the agents working collaboratively. This includes the entire flow: receiving a tenant inquiry, processing and routing the request, allocating tasks, and providing updates until task completion.

- Controlled Evaluation Using Query Dataset

A dataset of 100 realistic property owner queries was compiled based on commonly encountered issues in property management, such as repair requests, billing inquiries, and lease concerns. For each query, a benchmark answer was created manually to represent the expected high-quality response. These queries were used in a controlled evaluation to compare the multi-agent system against a single LLM (DeepSeek). The same question was posed to both systems, and responses were assessed using a structured rubric with five criteria (The detailed rubric is shown in Appendix):

Accuracy of Response – How correctly the system identifies the issue.

Completeness of Information – Whether the response fully addresses the query.

Clarity and Readability – Ease of understanding and language quality.

Response Time – How quickly the response is delivered.

Relevance of Context – How well the response fits the context of the query.

Stage 4: System Evaluation and Optimization

After testing, a detailed evaluation will be conducted to determine how well the system meets its original objectives and user expectations. The system will be reviewed in terms of both quantitative performance metrics and qualitative feedback from stakeholders.

- Performance Review

The system's response accuracy, task management efficiency, and user interaction quality will be analyzed in-depth. Particular attention will be given to areas where performance lagged, such as longer response times under load or misclassification of ambiguous queries.

- Stakeholder Feedback Collection

Feedback will be gathered from property managers, maintenance staff, and tenants through surveys, interviews, and usage logs. These insights will provide a user-centered perspective on:

System usability and accessibility.

The clarity and helpfulness of AI-generated responses.

Ease of tracking requests and receiving updates.

Overall satisfaction with task handling and resolution times.

- Optimization Process

Based on the collected feedback and performance data:

NLP models may be fine-tuned to better understand specific terminology used in property management.

The task allocation logic can be adjusted to improve efficiency during peak usage periods.

The communication agent's knowledge base may be expanded to cover more complex and nuanced queries.

User interface adjustments may be made to streamline interactions and improve accessibility.

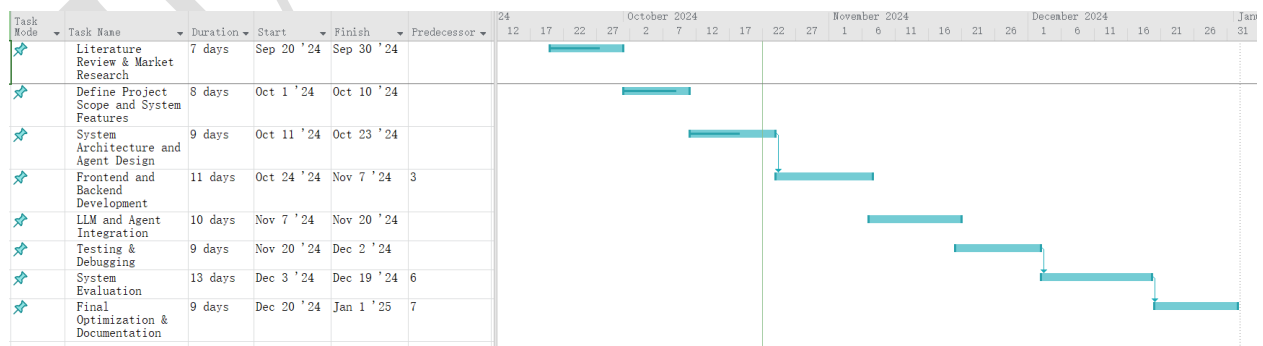
This final stage ensures that the system is not only technically sound but also practically effective and well-received by end users. The goal is to produce a robust, intelligent property management solution that is scalable, adaptable, and aligned with the evolving needs of modern property operations.

3.4 Project Schedule

The entire project will be done in 12 weeks' time. Follows is a detailed timeline of milestones and expected conclusions for each stage.

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Week	Task	Milestone	Expected Conclusion
1	Literature Review & Market Research	Complete market research and gather academic resources for related AI and multi-agent systems.	Identify gaps in current property management systems and establish a strong foundation for the project.
2	Define Project Scope and System Features	Finalize the specific features and objectives of the system, including key functions of each agent.	A well-defined project scope with clearly outlined system features and objectives.
3-4	System Architecture and Agent Design	Develop the system's architecture and agent communication protocols; flowcharts for agent interactions.	Finalized architecture, outlining how LLM and multi-agent systems will work together.
5-6	Frontend and Backend Development	Develop the frontend user interface and backend server with initial modules for communication agents.	A basic interactive system that can handle tenant inquiries and begin task management functions.
7-8	LLM and Agent Integration	Integrate the LLM for tenant communication and configure task allocation and monitoring agents.	Working system with automated tenant interactions and task allocation.
9-10	Testing & Debugging	Conduct unit tests for each module and fix any bugs in agent coordination, communication, and task scheduling.	Identified and fixed issues in communication, task management, and real-time monitoring.
11	System Evaluation	Evaluate system performance, including response time, tenant satisfaction, and task resolution efficiency.	Collected feedback from early users to improve system performance.
12	Final Optimization & Documentation	Optimize system functions and write final project documentation and user manual.	Completed system with detailed documentation ready for presentation and submission.



3.5 Key Milestones

- **System Architecture Completed:** A comprehensive architecture with clear agent workflows.

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- Functional Frontend and Backend: Initial system modules operational.
- Integrated LLM and Agents: Natural language processing and task automation active.
- Tested and Evaluated System: Performance metrics validated.
- Finalized System and Documentation: Ready for submission and deployment.

3.6 Software

- Frontend Framework: JavaScript; CSS; Socket.IO (Client); HTML.
- Backend Framework: JavaScript; Express.js; Socket.IO; DeepSeek; express-session; body-parser.
- LLM Integration: DeepSeek API for natural language understanding.
- Autonomous Agents: LangGraph.
- Database: Centralized relational database for storing and retrieving tenant and property data.

3.7 Hardware

Development Machine: Standard development laptops/desktops with the following specifications.

- ✓ 16GB RAM
- ✓ Multi-core processor (Intel i7 or equivalent)
- ✓ 512GB SSD storage

These machines will be used for development, testing, and running the application.

Cloud Resources: Cloud Computing Platforms (e.g., AWS, Google Cloud) for running the backend server and hosting the LLM integration.

Chapter 4 RESULTS AND DISCUSSION

4.1 Overview of Results

The Intelligent Property Management System (IPMS) was successfully developed and implemented, utilizing a robust technology stack. The backend was built with JavaScript, leveraging Node.js for server-side processing and Express.js to create a scalable, modular structure for handling user requests and managing tasks efficiently. Frontend was implemented using HTML, CSS and JavaScript to create the simplest and most responsive interface.

Some of the key technologies such as Socket.IO were included to facilitate real time interaction between tenants and property management. Real time updates for event handling included maintenance requests, chat interactions, and notifications that were handled in real time, thereby improving the overall user satisfaction as well as operations.

This chapter offers a detailed description of the system functional results, which confirms the proposed design objectives. It also analyzes the difficulties observed during the creation and the experimentation phases, and its strengths and weaknesses. In addition, it discusses how the system solves key issues in conventional property management systems, including slow communication, ineffective task management, and the requirement for improved tenant interaction in order to lay the groundwork for its implementation in real-life scenarios.

4.2 System Setup and User Guide

This subsection outlines the necessary steps to set up and run the Property Management System, as well as a brief description of the project structure and login credentials for testing purposes.

4.2.1 Project Structure

The directory structure of the Property Management System is organized as follows:

PropertyManagementSystem/

- |— node_modules/ # Installed Node.js dependencies
- |— public/ # Front-end assets (HTML, CSS, JavaScript)
 - | |— index.html # Landing page
 - | |— owner-dashboard.html # Dashboard interface for property owners
 - | |— staff-dashboard.html # Dashboard interface for property staff

|— server.js # Backend logic and Express server setup

|— package.json # Project metadata and listed dependencies

|— package-lock.json # Locked dependency versions for consistency

- node_modules/: Contains all installed Node.js packages required for the application to function.
- public/: Stores the HTML files for the landing page and user dashboards.
- server.js: The main backend script where the Express server is configured, routes are handled, and session logic is defined.
- package.json: Includes metadata such as the project name, version, and all declared dependencies.
- package-lock.json: Ensures consistent installations by locking specific versions of each dependency.

4.2.2 Installation Steps

Before running the application, ensure Node.js (with npm) is installed. It is the runtime environment required for executing the backend server and managing project dependencies. It can be downloaded from the official website: <https://nodejs.org>.

To set up the project on your local machine:

- Navigate to the Project Directory

Open a terminal and move into the project folder:

```
cd PropertyManagementSystem
```

- Initialize and Install Dependencies

Run the following commands to initialize the project and install required packages:

```
npm init -y
```

```
npm install express body-parser express-session
```

```
npm install socket.io
```

```
npm install openai
```

```
npm install node-fetch
```

```
npm install formdata-node
```

- Start the Server

After successful installation, launch the application by running:

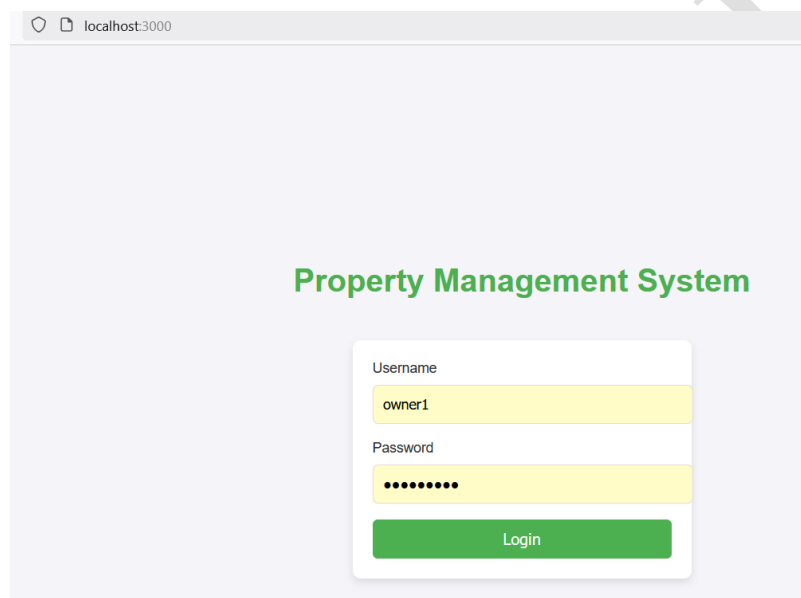
```
node server.js
```

```
Server is running on http://localhost:3000
A user connected
A user connected
A user disconnected
A user connected
A user disconnected
A user connected
```

- Access the Application

Once the server is running, open your browser and navigate to:

<http://localhost:3000/>



4.2.3 Login Credentials

To test the application, we can log in using the following sample user credentials:

- Owner 1
 - Username: owner1
 - Password: password1
- Staff 1
 - Username: staff1
 - Password: password2
- Owner 2
 - Username: owner2

- Password: 123

These credentials grant access to role-specific dashboards and allow users to interact with the system features such as managing properties, responding to queries, and handling requests.

4.3 Functional Achievements

4.3.1 Backend Implementation

The backend of the IPMS is the back end of the application that comprises of the engine that make the system to run efficiently. This activity is responsible for the verification of users, data storage, and also the distribution of work flows to guarantee how the system runs and the necessary enlargement. Below are the key results and features of the backend implementation:

- **Real-time Communication:** Using Socket.IO, the backend has strong real-time updates that are useful for the dynamic environment of the system. This functionality ensures that tenants and property management officials can easily communicate at first instance. Whether a tenant is reporting a repair issue, communicating with the property staff on a chat, or receiving updates on payment due or problems arising, all are conducted through Socket.IO with near real-time processing. This real-time interaction is one of the foundational principles of the system which has smooth and coherent user experience as its tenants and staff interact and respond accordingly.
- **Scalable Architecture:** Node.js and Express.js together make a very lightweight and pluggable backend architecture. Node.js for its part, which has non-blocking and event I/O model, ensures that the system can deal with multiple operations simultaneously. While, on the other hand, Express.js allows for building neat and easily modifiable routes, it improves the request-response cycle. This architecture was built with a view of expandability, a fact that make it very easy to expand the system in the future. For example, integrating third party APIs like payments gateways or other tools used in managing an additional property can easily be accomplished without necessarily redesigning the application. This modularity makes the system scalable so it can expand as the requirement of the property management ecosystem changes.
- **Task Management and Data Handling:** The backend is involved in organizing and distributing all the tasks that concern property maintenance, tenant's interactions, and payment procedures. It has the centralized database to enable it

to perform several functions effectively; it makes sure that all the requests are processed appropriately and at the same time. I have identified the following features as the most important for the system: The first one is the task allocation feature, which enables the system to give out maintenance tasks depending on the priority and availability of resources, urgency of the request among others.

- **Security and Privacy:** Security of data is a concern that receives a lot of attention when doing the backend development. Personal information of the tenants and properties is well protected and well handled with very tight security measures when it comes to data transfer. The system is to be compliant with privacy laws so access, and authorization control are to be very much in place. For user identification and session management, the express-session middleware is used to support secure and individualized experiences for the tenants and the property staff.

4.3.2 Frontend Features

The frontend of the Intelligent Property Management System (IPMS) was therefore, developed with simplicity, interactivity and user-friendly features in mind. Implemented with the use of HTML, CSS, and JavaScript, the structure of the windows and tabs offers elegant and easy to navigate environment for tenants and staff of the property management company. The following are the characteristics and outcomes of the frontend implementation:

- **Interactive Dashboards:** The property owners' and the staff's dashboards are the core of all the interactions occurring within the application. Owner Dashboard is a clear and convenient base in which users can find all the information about the property, create and view notices, make requests for maintenance work, and monitor the progress of current work. The Staff Dashboard which was developed with effectiveness in mind allows the staff to handle the repair requests, their tasks, and interact with the tenants. These dashboards are completely dynamic in nature and users can perform operations with a few clicks and without going to the next levels of the application.
- **AI Mode Integration:** One of the most important enhancements of the frontend is the inclusion of the AI Mode that enables the system's self-driven agents to engage tenants and handle property management activities. When in AI Mode, the system can reply to mails from tenants, update the status of the maintenance, notify the tenants, and even accept repair requests. This decreases the work load for

property staff and makes them to work on other more challenging tasks. From the user interface point of view, the AI Mode is perfectly embedded into the service with specific signals that inform the user whether the particular task is performed by an AI or a human, thus keeping the process clear and effective as to all customers.

- **Real-time Updates:** The communication within the system is real-time, which is why the frontend includes live updates for all property information. New property data, status of the maintenance requests, and new notification all of these changes are updated on the UI in real-time. In the frontend, the socket is set to listen to changes from the backend, and through the help of the Socket.IO package, it allows for changing of the content displayed without having to refresh the page. This real-time interaction also makes it possible for tenants and staff to always receive most recent information, increasing responsiveness and satisfaction. Users are always up-to-date with the new repair request, an update notice or even a payment reminder with little to no delay.
- **User-Centric Design:** The frontend which is designed in the manner provides smooth and easily navigable experience to the user. The layouts are responsive and the navigation straightforward which means even the less technical users will not have a problem with the system. The application of proper and easily understandable icons, drop down boxes, and real time blinking icon to indicate particular action or alert makes the application easier to use.
- **Customizable Notices and Alerts:** Besides the dashboards, the frontend of the new platform also allows property owners and staff to receive specific notices. Owners and managers can post public and personal announcements on the platform, which appear in the corresponding tabs of the staff's dashboard. This feature means that tenants are able to receive any message that may be relayed to them concerning the property, maintenance, payment, or otherwise, in the shortest time possible.

4.3.3 Module-Specific Results

Every module in the IPMS was developed addressing particular requirements in the property management process. The results of the implementation for each module are as follows:

- **Login Page:** The login page successfully differentiates between the owner and staff roles, ensuring that each user is directed to their respective dashboard with

the appropriate permissions and access levels. This kind of authentication makes sure that a user only has access to the data and features that is relevant to his position while at the same time it is user-friendly. The process of login is also very easy and while logging in, the user is given instructions on what to do in order to authenticate himself. When the users log in to our system they are automatically redirected to their respective custom dashboards making the user interface more efficient.

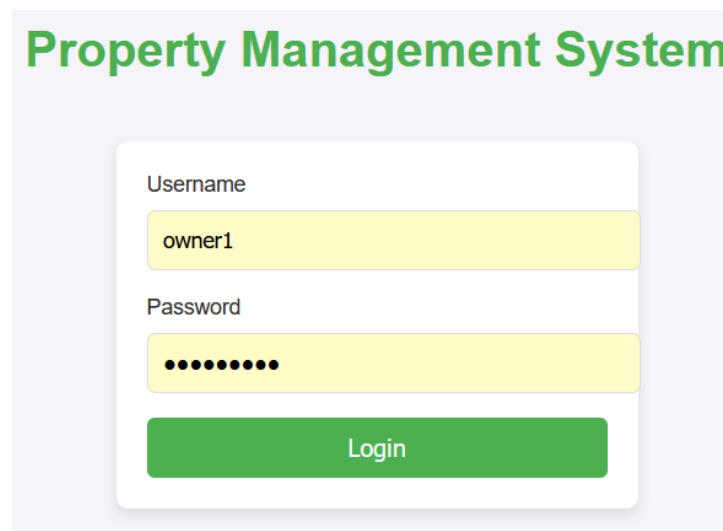


Figure 3. Interface of Login Page

- **Owner Dashboard:** The owner Dashboard allows the tenants to have a true picture of all the property information at any given time. It enables owners to view current information regarding their property such as current status of the properties, the maintenance requests pending for the property, and the personal messages. The system makes it possible for these updates to be processed instantly as facilitated by the real-time communication features. The owners can also request new repairs and use a convenient section to monitor their personal notifications. The concept of the dashboard is straightforward, and most of the essential information is provided in a rather uncomplicated manner, which will enable the owners to run their property without complications.

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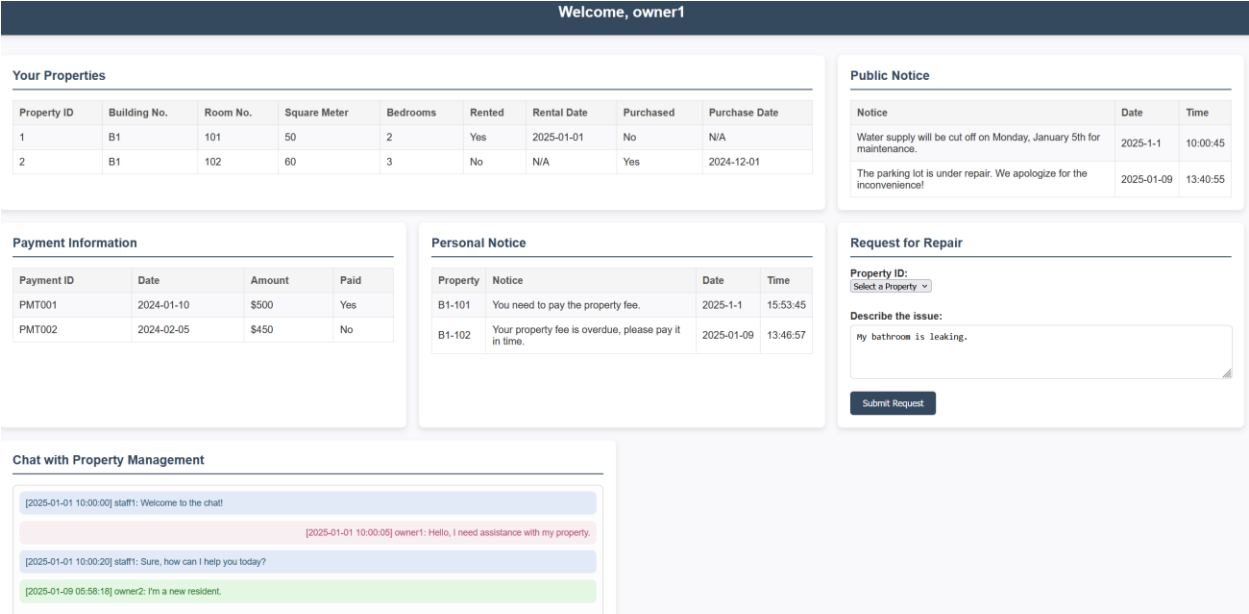


Figure 4. Interface of Owner Dashboard

- **Staff Dashboard:** The Staff dashboard is the central place for the management of property-related operations. Employees can easily monitor the maintenance requests, get the information on the properties, and send out notices. Notable among these is the incorporation of the AI Mode where management staff in the properties can have the autonomous agents handle the property’s day-to-day work including the handling of requests, and repair status updates. This relieves pressure on staffing since they will have less they worry about as other things are automatically handled. Communication and task management tools are also found in the dashboard to ensure a flow of the daily activities with greater efficiency.

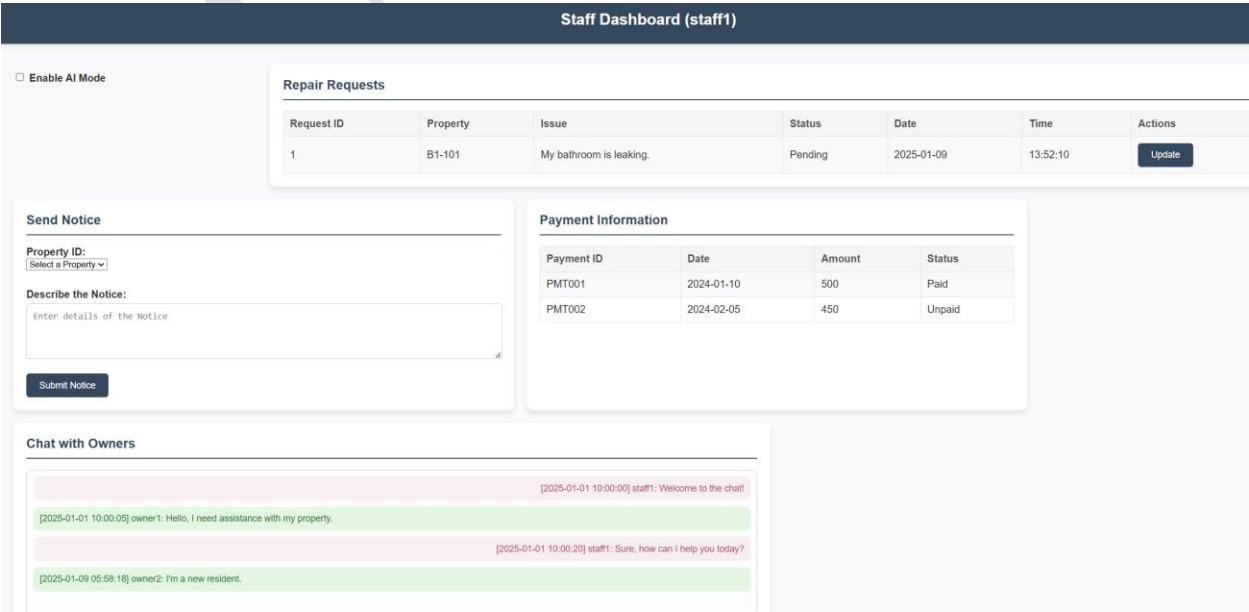


Figure 5. Interface of Staff Dashboard

- **Chat Module:** The Chat Module enables convenient and effective real-time communication between owners and the staff. It encompasses the role-based message identification which assists the users to identify who is communicating with them. Messages that were sent by the user himself or herself are marked in red, staff messages are marked in blue and messages from other owners are marked in green. Messages that were sent by the agent are marked in green. This clear separation enhances the content organization thus envisaging easy following of conversations by the users. The chat system is seamlessly embedded within the IPMS, which ensures that all the communications are in real time, especially with issues to do with repairs, property notices as well as tenant concerns.

Chat with Property Management

[2025-01-01 10:00:00] staff1: Welcome to the chat!

[2025-01-01 10:00:05] owner1: Hello, I need assistance with my property.

[2025-01-01 10:00:20] staff1: Sure, how can I help you today?

[2025-04-10 02:48:31] owner1: Are pets allowed in the building?

[2025-04-10 02:48:30] Ai: Pets are generally not allowed in the building unless they are service animals or emotional support animals with proper documentation. Please check with the building management for specific policies and any required approvals before bringing a pet onto the premises.

Type your message here...

Send

Figure 6. Panel of Owner Chat

Chat with Owners

[2025-01-01 10:00:00] staff1: Welcome to the chat!

[2025-01-01 10:00:05] owner1: Hello, I need assistance with my property.

[2025-01-01 10:00:20] staff1: Sure, how can I help you today?

[2025-04-10 02:48:31] owner1: Are pets allowed in the building?

[2025-04-10 02:48:30] Ai: Pets are generally not allowed in the building unless they are service animals or emotional support animals with proper documentation. Please check with the building management for specific policies and any required approvals before bringing a pet onto the premises.

Type your message here...

Send

Figure 7. Panel of Staff Chat

- **Notice (Public):** The Public Notice module enables the staff to post the notices to all the property owners at once. If staff post a notice and do not choose a property, the system defaults the notice to be a public notice. All owners will receive this notice in the Public Notice tab. This feature is particularly helpful for any general notifications some of which include maintenance of the compound, changes in policies, or an urgent message to every tenant. The interface is designed to be straightforward: staff can easily construct and disseminate the notice and owners can promptly view and respond to the notice. This makes it possible for communication to take place, and it reaches all the relevant stakeholders within the shortest time possible.

Send Notice Staff Dashboard

Property ID:
Select a Property ▼

Describe the Notice:
The parking lot is under repair. We apologize for the inconvenience!

Submit Notice

Public Notice Owner Dashboard

Notice	Date	Time
Water supply will be cut off on Monday, January 5th for maintenance.	2025-1-1	10:00:45
The parking lot is under repair. We apologize for the inconvenience!	2025-01-09	13:40:55

Figure 8. Panel of Send Notice and Public Notice

- **Notice (Personal):** The Personal Notice module is intended to notify certain property owners with particular notices. When the staff uses the drop down list to send a notice to a specific property, the owner of that property finds the notice in their Personal Notice tab. Such a mass-targeted approach is appropriate for such items as repairs and other works, lease renewal, or payment reminder for specific properties. It is easy for staff – after choosing the property, the system guarantees that the notice will be delivered to the proper owner. This makes personal and property-specific communication precise, relevant, and systematic to the extent that management of property affairs becomes more convenient to the owners and other staff.

Send Notice Staff Dashboard

Property ID:
B1-102
Select a Property ▼

Describe the Notice:
The parking lot is under repair. We apologize for the inconvenience!

Submit Notice

Personal Notice Owner Dashboard

Property	Notice	Date	Time
B1-101	You need to pay the property fee.	2025-1-1	15:53:45
B1-102	Your property fee is overdue, please pay it in time.	2025-01-09	13:46:57

Figure 9. Panel of Send Notice and Personal Notice

4.3.4 LLM Integration

The integration of a Large Language Model (LLM) significantly enhanced the system's ability to understand and process natural language, elevating the user experience and streamlining tenant interactions:

- **Natural Language Understanding (NLU):** The LLM allows the system to understand and respond appropriately to all sorts of tenant questions and requests for repairs. High NLU possibilities allow understanding user's intentions, defining key entities (apartment number, problems with appliances, etc.) and offering the proper response to meet the tenants' needs.
- **Contextual Conversations:** The LLM keeps track of the conversation context from one interaction to another, thereby making multi-turn, coherent and on-topic conversations possible. This means that tenants can have more fluid conversations that are not interrupted by the need to repeat information which also makes the process much easier to use. In the same thread of conversation, the system can recall the previous messages, thus improving the quality of the replies and minimizing the number of instances the tenants would need to repeat themselves.

4.4 Comparative Experiment: Multi-Agent System vs. DeepSeek

To evaluate the performance of the proposed multi-agent system, a comprehensive comparative experiment was conducted against DeepSeek, a well-known AI model. The goal was to assess each system's ability to handle common property management queries effectively, with a focus on accuracy, completeness, clarity, contextual relevance, and response efficiency.

Dataset Design:

A total of 100 questions were generated using a language model (DeepSeek) and reviewed for quality and realism. These questions were grouped into five essential categories of property management:

- Policies (e.g., pets, smoking rules, guest limits)
- Contracts (e.g., subletting, termination terms, lease duration)
- General Inquiries (e.g., account access, office hours, contact info)
- Maintenance (e.g., emergency repairs, contractor rules)
- Payments (e.g., due dates, late fees, payment methods)

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Each category included 20 questions, ensuring a diverse and balanced evaluation. For consistency and objective scoring, a standard answer was manually crafted for each question by referencing real-world property management policies and best practices. These served as the ground truth for evaluating response quality.

Evaluation Criteria

Each answer—whether generated by the Multi-Agent system or DeepSeek—was scored based on the following five dimensions:

Dimension	Description	Max Points
Accuracy of Response	Degree to which the response matches the facts and intent of the standard answer.	50
Completeness	Coverage of all required elements, including both explicit and implied details.	20
Clarity & Readability	Ease of understanding, grammar, structure, and conciseness.	15
Response Time	Speed at which the system delivers an answer (measured manually and averaged).	10
Contextual Relevance	How well the system maintains context and stays on-topic.	5
Total		100

Each system's response to all 100 questions was manually scored, then average scores were computed for comparative analysis.

Example Evaluation

Sample Question: "Are pets allowed in the building?"

Standard Answer: "Pets are allowed with prior approval and a pet deposit."

Multi-Agent Response: " Pets are permitted in the building with prior approval. To bring a pet, you must: Complete official registration with property management; Pay the required pet deposit.

Please note: All pets must be well-behaved and not disturb other residents; Property management reserves the right to revoke pet privileges for any behavioral issues; Additional fees or restrictions may apply for certain breeds/sizes"

Accuracy: 45/50 (fully aligned with standard answer)

Completeness: 18/20 (adds behavioral policy without losing essential details)

Clarity: 15/15

Response Time: 7/10

Context Relevance: 5/5

Total Score: 90/100

DeepSeek Response: " Pets are generally allowed in the building, but there may be specific rules or restrictions depending on the property's policies. Common guidelines include:

Leash & Supervision: Pets should be on a leash or in a carrier in common areas.

Noise & Behavior: Pets must not disturb other residents (excessive barking, aggression, etc.).

Breed/Size Restrictions: Some buildings have limits on pet size or prohibit certain breeds.

Pet Fees/Deposits: There may be additional fees or a pet deposit required.

Waste Cleanup: Owners must clean up after their pets.

... " (quite long)

Accuracy: 40/50 (some drift from the specific property policy)

Completeness: 18/20 (covers key ideas, but with generic phrasing)

Clarity: 12/15 (verbose with some redundancy)

Response Time: 9/10

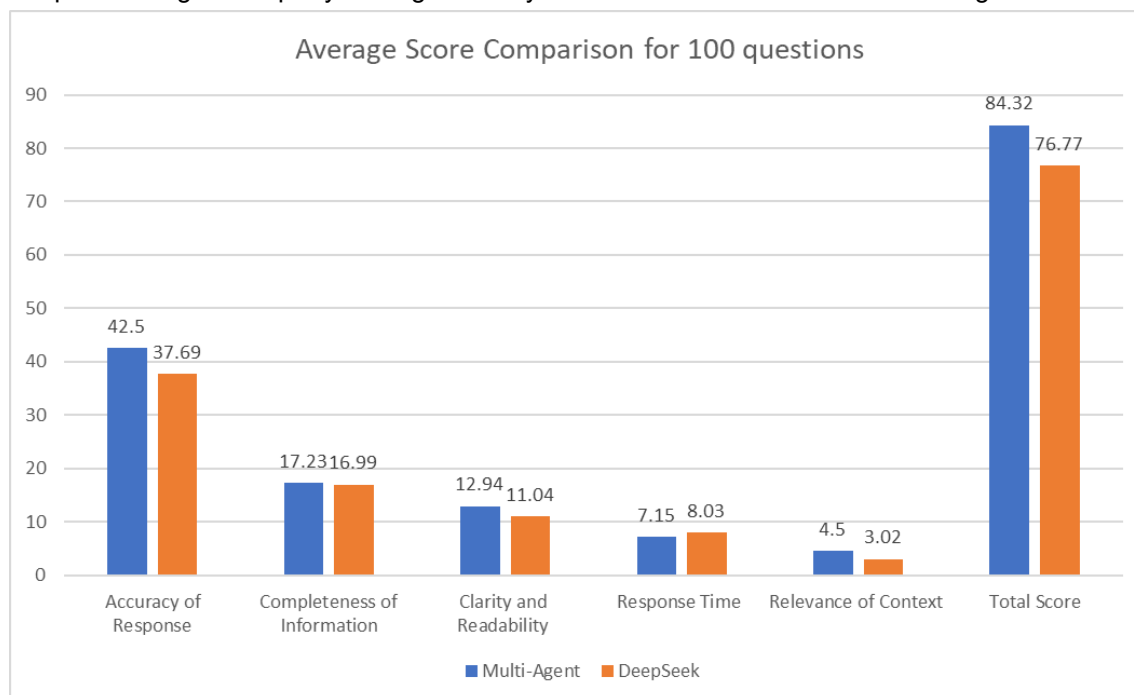
Context Relevance: 3/5

Total Score: 82/100

Overall Results

The table below presents the average score across all 100 questions in each of the five evaluation categories:

Category	Multi-Agent System	DeepSeek
Accuracy of Response	42.5 / 50	37.69 / 50
Completeness	17.23 / 20	16.99 / 20
Clarity & Readability	12.94 / 15	11.04 / 15
Response Time	7.14 / 10	8.03 / 10
Context Relevance	4.5 / 5	3.02 / 5
Total Score	84.32 / 100	76.77 / 100



The experiment highlights several strengths of the multi-agent system, particularly in its ability to deliver responses that are precise, policy-aligned, and tailored to the context of property management. One of the key advantages lies in task specialization. By assigning specific responsibilities to individual agents—for example, handling contracts, processing payments, or responding to policy-related inquiries—the system consistently generated answers that reflected organizational rules and procedures more accurately than a general-purpose model could. This structure reduced ambiguity and allowed for more relevant, detailed responses.

Another notable strength of the multi-agent system is its context management capability. It demonstrated strong continuity across multi-turn interactions, effectively maintaining conversation memory and adapting its responses to prior queries. This was particularly valuable in cases where users followed up with related questions, as the system was able to recall and build upon earlier answers. In addition, the system produced clear and concise output, avoiding the use of overly generic or verbose explanations. This focus on clarity and readability contributed to better user comprehension and satisfaction.

In contrast, DeepSeek exhibited certain limitations that affected its performance in this domain-specific context. A key issue was generalization drift—its tendency to produce answers that, while factually plausible, were overly broad and not aligned with specific property policies or procedures. As a result, its responses sometimes lacked the precision necessary for effective tenant communication. Furthermore, DeepSeek showed weakness in context management, occasionally misinterpreting the intent of follow-up

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questions or shifting off-topic. These lapses made it less reliable in handling sequences of related inquiries. Another drawback was its verbosity: DeepSeek often included excessive detail or repetitive phrasing, which diluted the core information and reduced overall clarity.

While DeepSeek did outperform the multi-agent system in terms of response time, this advantage came with a tradeoff. Its faster replies were often less accurate or relevant, especially in scenarios that demanded nuanced understanding or policy compliance. In contrast, the slightly slower responses from the multi-agent system were more contextually grounded and better aligned with the organization's operational needs. Ultimately, the added processing time was justified by the higher quality of information delivered.

These findings support the view that domain-specific, multi-agent systems are better suited for complex customer service tasks in property management. Their ability to interpret and apply internal policies, manage contextual continuity, and communicate clearly makes them a more effective tool for enhancing operational efficiency and user experience.

4.5 Challenges Encountered

4.5.1 Key Technical Challenges

One of the most complicated tasks was the ability to track a conversation history of the client and recall history of the conversation with this client in particular. Due to the constant flow of tenant requests, the system had to retain context to give the best answers. This was resource hogging and had to manage the state and memory of the system. The concept of maintaining a mechanism that would enable the tracking of the flow of conversation between the agents while not weighing heavily on the functionality of the system was a big challenge in the design process.

Performance issues were observed under stress testing especially concerning concurrency, that is number of requests. Although the system could easily handle a moderate load, lag in response times was observed at times of high traffic. This brought out the need to do more optimization such as improving the scalability of the base system, improving the speed of algorithms used in the system, and use techniques such as load balancing in order to achieve real time processing during busy hours.

Confidentiality was a highly significant factor because the system would entail personal

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information of tenants, and details about repair and maintenance. The basic principle of an encrypted connection, secure user sessions, and adherence to the latest data protection standards (such as GDPR or CCPA) were vital for implementing the platform's security. However, these security measures have been known to have consumed a lot of development time and resources to complete effectively. Another preventive measure that was taken care of was that it shouldn't hamper performance when security protocols were being implemented.

4.5.2 LLM-Specific Challenges

- **Language Model Training:** Adapting LLM to property-specific language and laws was a challenging and a long process. However, general-purpose LLMs are very effective, but they have a problem when it comes to specific professional languages or specific topics. To overcome this, the model had to be trained on specialized datasets that include terms like property management, legal terminologies of the region and other phrasal terminologies of property management. In this fine-tuning, there was an intricate loop of checking and rechecking to arrive at the perfect model that can adequately respond to queries of the context and meet the needs of the property management context.
- **Multilingual Context:** The major issues involved in maintaining the conversation context across multiple languages were identified. Although the LLM could accommodate several languages, the system encountered the problem in answering sophisticated or localized queries that contained cultural or local cues. Adapting to an idiomatic request while keeping the model aware of the conversation's context was even challenging when transitioning from one language to another. It had to be designed to be able to store and recall information when users spoke in different languages in the course of a particular conversation. This entailed constant practice to avoid the change of the language to cause the respondent to lose track of the questions or give ambiguous answers.
- **Bias and Fairness:** Like any other machine learning model, preconception detection and reduction in the LLM was a continuous process. The model was intended to be bias free and equal for all, but they can be introduced during training data or some underlying social structure. Many pilots and iterations were needed to uncover possible sources of bias regarding race, gender, or socioeconomic status. This included performing checks to determine how well the model performs on a large variety of queries as well as checking if the replies given were bias free

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and fair. Mitigating these issues was possible by several modifications to the training process and constant vigilance to achieve fairness of the model's outputs.

4.6 Discussion of Outcomes

4.6.1 Successes

The implementation of the Intelligent Property Management System (IPMS) has achieved most of its key goals and has provided enhancements in several segments of property management. Notable successes include:

- **Operational Efficiency:** One of the standout successes was the automation of routine tasks such as handling repair requests, maintenance scheduling, and tenant inquiries. This meant that staff was relieved some of the burden and was free to attend to other activities which are perhaps more demanding or which add value to the organization. The automation of request processing and responses to them also contributed to the efficiency of the work carried out within the framework of property management.
- **User Satisfaction:** AI enabled interactions in the form of real time communication played a role in enhancing tenant satisfaction. Tenants were going to be able to engage the system at their own convenience and get responses to their questions and/or commands within shortest possible time. Besides improving the usability and giving users the feeling 'being listened to', this responsiveness created a positive culture for reliability and transparency. Also, the means to monitor the status of repair requests and get updates without having to wait to receive them manually introduced a smoother, proactive system whose functions greatly improved tenant satisfaction and involvement.
- **Scalability:** It was also identified that the system was designed with modularity as its core strength; this means that it can be enhanced in future without much complications on the existing design. Whenever new requirements for the features or functionalities are required—be it for supporting new languages, or integrating the system with other services, or for adding new capabilities using AI—the scale up can be done without necessarily requiring the system to be redesigned. This flexibility guarantees that the IPMS will be able to adapt to the new changes in business and technologies thus being a long term solution in property management.

4.6.2 Limitations

While the IPMS achieved considerable success, there were several limitations that must

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be addressed in future iterations:

- **High Dependency on LLM:** This is because the system's NLU and NLG heavily depend on the Large Language Model (LLM) that is at the core of the system. Though the LLM is efficient in most of the scenarios, it may show poor performance in specific cases, including searching with specialized terms, ambiguous queries, or complicated tenant queries. As the system increases in size and sophistication, it will be important to keep LLM strong and prepared for a large number of different inputs. As well, any weakness in the LLM's capability to understand or produce answers might entail misinterpretation or generation of false information thereby resulting to a low tenant satisfaction level.
- **Real-Time Stress Handling:** At one point, during periods of high usage or high loads, the system would sometimes fail to serve multiple high-volume requests concurrently. While the system demonstrated good performance in most typical scenarios, there is still a lot of work to be done to make it possible to achieve high-scale real-life interactions without degradation of the performance. It may mean refining systems structures, increasing the effectiveness of organizational load management approach, and consolidating the algorithms to lessen subject processing time under a large number of requests. Solving these problems would enhance the system's capacity for delivering fast and responsive service all the time, including periods of high traffic.
- **Limited Training Data for Multilingual Capabilities:** The translation function in the system is functional but it can be very much enhanced by a more diverse and extensive dataset. While the system is multilingual it can still improve its operation when it comes to region-specific dialects and local terminology as well as culturally sensitive dialogues. The training data currently used can be limited in the amount of linguistic variation and the level of language complexity adequate in all the target areas. Increasing the number of languages spoken, different variants, and regionalisms detectable by the system can increase the effectiveness of its responses to interactions with tenants on a global level.

Chapter 5 CONCLUSION

5.1 Relevance to the objectives

In the light of these objectives, the introduction of the Intelligent Property Management System (IPMS) was achieved as planned. These objectives were primarily focused on making it easier to operate as well as making communication between staff and tenants easier, and to also ensure that the system could support the organizations growth.

- **Operational Efficiency:** The IPMS has greatly enhanced property management tasks and responsibilities by providing solutions for simple functions like submitting repair requests, making inquiries to the tenant or even scheduling for maintenance. These automations have helped in easing the working pressure on staff so that they can work on other important tasks, as well as helped in enhancing the working operations within the property management team. The end effect is a more efficient and less expensive method of operation, thus improving the organization's profitability.
- **Tenant-Staff Communication:** The system has made communication between tenants and property management more efficient and faster. Thanks to the integration of real-time communication features and artificial intelligence, the IPMS enables the tenants to make requests and follow up requests and feedback, resolve issues and specify the outcomes without time lags inherent to conventional communication means. This has in turn contributed to enhancement of tenant satisfaction and interaction with the buildings they occupy.
- **Scalability and Adaptability:** The flexibility of the IPMS is achieved through the modular design of the system hence the system will be able to accommodate the future needs of the organization. For extra languages, new options, or connection to other technologies, such changes do not require a major overhaul of the system. This flexibility will enable the IPMS to be able to address the needs of property management as the business expands and technologies change.

5.2 Recommendation

While the IPMS has demonstrated strong success in meeting its core objectives, there are several areas where further development could enhance its performance and address existing limitations:

- **Improvement of LLM Accuracy and Efficiency:** Due to the high dependence on the LLM for natural language processing, some additional tuning is advised to enhance the performance of the solution because of its weak performance with

specific terminology, ambiguous queries, and complicated tenant requests. Incorporating more diverse, domain-specific training data and refining the model's understanding of property management concepts will increase the LLM's robustness and reliability.

- **Enhanced Performance Under High Load:** In order to deal with such issues that have been observed during stress testing, it is suggested to improve the system for real-time work with high request frequencies. This could include improving the base architecture, improving the algorithms, and putting into practice even superior load-balancing techniques in the course of peak load.
- **Expansion of Multilingual Capabilities:** Thus, to serve the versatile tenant population, there is a need to enhance the system's linguistic versatility. The expansion of the multilingual datasets will also enhance the system's performance in addressing the regionally specified dialects and culturally related language peculiarities. This expansion would guarantee that different tenants across geographic locations get proper, relevant and localized responses in their preferred languages thus increasing the versatility of the system.
- **Continuous Monitoring for Bias and Fairness:** To maintain the fairness and inclusivity of the system, it is essential to conduct ongoing audits and testing to identify and mitigate any biases in the LLM. This ensures that the system continues to treat all tenants equitably, regardless of their background or personal characteristics.

In conclusion, the IPMS has proven to be an effective and valuable tool in modern property management, achieving its core objectives and improving both operational efficiency and tenant satisfaction. By addressing the recommendations above, the system can continue to evolve and meet the growing demands of property managers and tenants alike.

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Appendix

A. Rubric to evaluate the multi-agents

1. Accuracy of Response (50 points)

- 45-50 points: The response is fully accurate, aligns perfectly with the standard answer, and provides detailed and relevant information.
- 35-44 points: The response is mostly accurate but may contain minor factual errors or slight deviations from the standard answer.
- 25-34 points: The response is partially accurate but misses key details or includes noticeable inaccuracies.
- 10-24 points: The response shows limited accuracy, with significant factual errors or missing important information.
- 0-9 points: The response is completely inaccurate or irrelevant.

2. Completeness of Information (20 points)

- 18-20 points: The response covers all aspects of the question thoroughly, addressing both explicit and implicit details.
- 14-17 points: The response covers most of the key details but lacks some minor information.
- 10-13 points: The response addresses only part of the question, missing some important details.
- 5-9 points: The response is incomplete, covering only a small portion of the required information.
- 0-4 points: The response is completely incomplete or irrelevant.

3. Clarity and Readability (15 points)

- 13-15 points: The response is clear, concise, and easy to understand. It uses proper grammar, spelling, and sentence structure.
- 10-12 points: The response is mostly clear but contains minor grammatical or readability issues.
- 7-9 points: The response is somewhat clear but may include awkward phrasing or unclear sentences.
- 4-6 points: The response is difficult to understand due to poor grammar, unclear structure, or confusing phrasing.
- 0-3 points: The response is incoherent or incomprehensible.

4. Response Time (10 points)

- 9-10 points: The AI responds in under 2 seconds.
- 7-8 points: The AI responds in 2-4 seconds.
- 5-6 points: The AI responds in 4-6 seconds.
- 3-4 points: The AI responds in 6-10 seconds.
- 0-2 points: The AI takes longer than 10 seconds to respond or times out.

5. Relevance of Context (5 points)

- 5 points: The response maintains full contextual relevance, accurately following the conversation thread.
- 4 points: The response is mostly contextually relevant but may slightly deviate from the conversation.
- 3 points: The response is moderately relevant but may show noticeable context drift.
- 2 points: The response has weak contextual relevance, partially misinterpreting the conversation.
- 0-1 point: The response is completely irrelevant or off-topic.

Total Score: 100 points

Final Evaluation:

- 90-100 points: Excellent performance.
- 75-89 points: Good performance with minor issues.
- 50-74 points: Average performance, with room for improvement.
- 25-49 points: Below average, with significant inaccuracies or issues.
- 0-24 points: Poor performance or failure.

B. Evaluation Questions

Category	Question	Standard Answer
Policies	Are pets allowed in the building?	Pets are allowed with prior approval and a pet deposit.
Contracts	Are subletting agreements allowed?	Subletting agreements are not allowed unless explicitly approved by the management.
General Inquiries	Can I access my account online?	Yes, you can access your account through our online portal.
Contracts	Can I add another tenant to my contract?	Adding a tenant requires landlord approval and an updated lease agreement.
Maintenance	Can I hire my own contractor for repairs?	Tenants must use authorized contractors for repairs.
Policies	Can I modify the interior of my property?	Interior modifications require written approval from management.
Payments	Can I pay my rent using a credit card?	Yes, rent can be paid via credit card through the online portal.
Payments	How can I set up automatic payments?	Automatic payments can be set up through the tenant portal.
General Inquiries	How do I contact the emergency services?	For emergencies, dial 999. For property-related emergencies, call our 24-hour hotline 0852-36138727.
Contracts	How do I renew my lease?	Lease renewal procedures vary. Contact management 60 days before lease expiration.
Maintenance	How do I request a repair for my unit?	Repair requests can be submitted via the tenant portal or by calling the maintenance team.
Payments	Is there a late payment fee?	Yes, a late payment fee of \$50 is charged if rent is overdue by more than 5 days.
General Inquiries	What are the office hours of the property management?	Office hours are Monday to Friday, 9 AM to 5 PM.
Policies	What are the rules regarding visitor parking?	Visitor parking rules are specified in the lease agreement and vary by property.
Contracts	What happens if I break my lease early?	Breaking a lease early may result in penalties or forfeiture of the security deposit.
Policies	What is the policy on noise complaints?	Noise complaints should be reported to management, and repeated violations may lead to fines.
Maintenance	What is the typical response time for maintenance issues?	Maintenance response times depend on the issue but typically range from 24 to 48 hours.
Maintenance	What should I do if my heating system stops working?	If your heating stops working, report it via the tenant portal or call maintenance.

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Payments	When is the rent due each month?	Rent is due on the 1st of each month, with a grace period until the 5th.
General Inquiries	Where can I find the property guidelines?	Property guidelines are available on the tenant portal or in your lease agreement.
Policies	Are smoking and vaping allowed in the building?	Smoking and vaping are strictly prohibited in all indoor areas of the property.
Policies	Can I install a satellite dish on the property?	Satellite dish installation requires written approval and must follow guidelines.
Contracts	Can I break the lease if I get a new job in another city?	Lease termination due to relocation is subject to early termination clauses.
Maintenance	How often is pest control service conducted?	Pest control is scheduled quarterly, or upon request if issues arise.
Payments	Can I split rent payments with my roommate?	Yes, but the full rent must still be paid on time; roommates can divide it privately.
Policies	Is there a curfew for shared common areas?	Quiet hours are from 10 PM to 7 AM in shared spaces.
Contracts	Do I need renters insurance?	Renters insurance is strongly recommended and may be required per lease terms.
General Inquiries	Are there any community events for tenants?	Yes, community events are announced via email or on the tenant portal.
Policies	Can I list my unit on Airbnb?	Short-term rentals such as Airbnb are prohibited without management approval.
Maintenance	Who is responsible for replacing lightbulbs?	Tenants are responsible for replacing lightbulbs in their units.
Payments	Can I get a receipt for rent payments?	Yes, receipts are automatically generated in the tenant portal.
General Inquiries	Is there a gym in the building?	Yes, a gym is available for resident use from 6 AM to 10 PM daily.
General Inquiries	Are there laundry facilities on-site?	Yes, shared laundry rooms are available on each floor.
Maintenance	How do I report a pest problem?	Pest problems should be reported through the portal or by calling the office.
Contracts	Can I terminate the lease if I lose my job?	Job loss does not automatically release you from the lease; contact management.
Policies	Can I host parties in my unit?	Small gatherings are allowed, but must comply with noise and occupancy rules.
General Inquiries	Is there bicycle storage available?	Yes, bike racks and storage rooms are available upon request.
Contracts	What happens when the lease ends?	You may choose to renew or move out; notice is required 30-60 days in advance.
Payments	Is there a security deposit refund policy?	Security deposits are refunded within 30 days of move-out, pending inspection.
Maintenance	Are smoke detectors regularly checked?	Yes, they are inspected annually and tenants should report any issues immediately.
General Inquiries	How do I update my contact information?	Update your info via the tenant portal or by contacting the office.
Policies	Are BBQs allowed on balconies?	BBQs are not allowed on balconies due to fire regulations.
Contracts	Can I extend my lease for a shorter term (e.g., 3 months)?	Short-term extensions may be available; contact management for options.
Payments	Can I change my payment method mid-lease?	Yes, you can update payment methods in the tenant portal at any time.

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Maintenance	Do you provide seasonal maintenance checks?	Yes, seasonal inspections are conducted, including HVAC and plumbing systems.
General Inquiries	Can I receive packages at the property?	Yes, packages are accepted at the front desk or delivered to parcel lockers.
Policies	Is recycling mandatory?	Yes, all residents are required to follow recycling procedures outlined in the guide.
Contracts	What documents are needed to sign a lease?	A valid ID, proof of income, and completed application form are required.
Maintenance	Who do I call for an after-hours plumbing emergency?	Call the 24/7 emergency maintenance hotline provided in your welcome packet.
General Inquiries	Can I tour the unit before signing the lease?	Yes, unit tours can be scheduled through the property management office.
Policies	Are smoking and vaping allowed in the building?	Smoking and vaping are not permitted inside the building; designated outdoor areas are provided.
Policies	Can I hang things on the walls?	Yes, but you are responsible for patching holes before move-out.
Policies	Is there a restriction on the number of guests I can have?	Extended guests staying over 7 consecutive days require management notification.
Policies	Are outdoor grills allowed?	Grills are not permitted on balconies or patios for safety reasons.
Policies	Can I keep exotic pets?	Exotic pets are not allowed due to safety and insurance policies.
Policies	Can I plant a garden outside my unit?	Personal gardening is allowed only in designated community garden areas.
Policies	What is the parking policy for multiple vehicles?	Each unit is allowed one reserved spot; additional parking is subject to availability.
Policies	Can I store items in the hallway?	Hallways must be kept clear of personal items for fire safety.
Policies	Is there a guest policy for using amenities?	Guests must be accompanied by a resident when using amenities.
Policies	Are children allowed in the gym?	Children under 14 must be supervised by an adult in the gym.
Contracts	Can I change the lease start date?	Changes to lease start dates require prior management approval.
Contracts	What is the lease duration?	Standard leases are 12 months unless otherwise specified.
Contracts	Can I request a month-to-month lease?	Month-to-month options may be available after the initial lease term.
Contracts	What is the penalty for breaking the lease?	The penalty is typically one month's rent plus forfeiture of the security deposit.
Contracts	Can I transfer my lease to someone else?	Lease transfers require management approval and a background check of the new tenant.
Contracts	What if one roommate wants to move out?	The remaining tenant must reapply or find a new co-tenant subject to approval.
Contracts	Do I need a guarantor?	A guarantor may be required if your income doesn't meet the minimum threshold.
Contracts	How do I terminate my lease at the end of the term?	Submit a written notice 60 days before lease expiration.
Contracts	Can I request a lease in digital format?	Yes, digital leases are available and legally binding.
Contracts	What are the consequences of non-renewal without notice?	Failure to notify may result in automatic lease renewal or additional charges.

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Payments	Can I pay rent in cash?	Cash payments are not accepted for security reasons.
Payments	Do you offer rent payment assistance plans?	Payment plans may be arranged for tenants facing financial hardship.
Payments	Is there a charge for returned payments?	Yes, a \$35 fee applies to returned or bounced payments.
Payments	Can I prepay rent?	Yes, rent can be prepaid in advance with management confirmation.
Payments	How are utility charges calculated?	Utility charges are based on individual or shared meter readings.
Payments	Are utilities included in the rent?	Utilities may or may not be included; check your lease for details.
Payments	What payment platforms do you support?	Payments can be made via the tenant portal, bank transfer, or credit/debit card.
Payments	Can I get a copy of my payment history?	Yes, payment history is available through your online account.
Payments	What happens if my payment is declined?	You will be notified to submit a new payment method within 3 business days.
Payments	Do you offer discounts for early rent payment?	No standard discounts are offered, but promotional incentives may apply.
Maintenance	Are emergency repairs available on weekends?	Yes, emergency maintenance is available 24/7 including weekends.
Maintenance	How do I track the status of a repair request?	You can track repair progress through your online portal account.
Maintenance	Do I have to be present for maintenance work?	Not unless you prefer to be; staff can access your unit with notice.
Maintenance	How do I report mold or water damage?	Immediately submit a maintenance request and notify the office.
Maintenance	Are HVAC filters replaced by the property?	Yes, filters are replaced quarterly by our maintenance team.
Maintenance	Can I be charged for maintenance?	Routine maintenance is free; tenant-caused damage may incur fees.
Maintenance	Can I paint the walls?	Only with written permission; you may be required to repaint before move-out.
Maintenance	How often is landscaping done?	Landscaping is done biweekly or as weather permits.
Maintenance	What is considered an emergency repair?	Plumbing leaks, power outages, or broken locks are emergency issues.
Maintenance	How are common areas maintained?	Common areas are cleaned and inspected daily by janitorial staff.
General Inquiries	Is there a community newsletter?	Yes, monthly newsletters are emailed and posted in the tenant portal.
General Inquiries	Do you offer moving assistance or carts?	Yes, carts and dollies are available upon request at the front office.
General Inquiries	Is there a recycling program?	Yes, all residents are encouraged to follow our recycling program guidelines.
General Inquiries	Can I get a copy of the lease agreement?	Yes, a copy is available on the portal or by contacting the office.
General Inquiries	Is the property wheelchair accessible?	Yes, all buildings meet ADA accessibility standards.
General Inquiries	Are there elevators in the building?	Yes, elevators are available in all multi-level buildings.

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General Inquiries	Is public transport easily accessible?	Yes, there are bus stops and train stations within walking distance.
General Inquiries	Are security cameras installed?	Yes, cameras are located in all entrances, exits, and common areas.
General Inquiries	How do I book amenities like the clubhouse?	Use the tenant portal to reserve amenities or contact the office.
General Inquiries	Are smoke-free units available?	Yes, we offer smoke-free buildings upon request.

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